Icarus rev 0.5 — dual-channel triple-shutdown steering torque-overlay unit

CH-A (F280049C-Q1) computes the overlay, CH-B (F280049C-Q1) monitors it, CH-C is an analog guard with no software.
Overlay reaches the EPS only through relays K1 AND K2; each coil needs CH-A high side + CH-B low side + CH-C guard FET.
With no power or any enable missing, NC contacts connect the OEM torque sensor straight to the EPS.
Safety concept: ../docs/04_TECHNICAL_SAFETY_CONCEPT.md. Values marked TBD-DATA wait for logger measurements.

power

Power: input protection and two independent supply chains

File: power.kicad_sch

ch_a

CH-A: F280049C-Q1 control (computes and drives the overlay)

File: ch_a.kicad_sch

ch_b

CH-B: F280049C-Q1 monitor (checks; owns the low-side coil enable)

File: ch_b.kicad_sch

overlay

Overlay path (CH-A analog, +5V0_A)

File: overlay.kicad_sch

guard

CH-C analog guard (no software, +5V0_B)

File: guard.kicad_sch

relays

Signal relays K1/K2 (cascaded) and cross-driven coils

File: relays.kicad_sch

io

Vehicle connector, CAN bus, inputs and monitors

File: io.kicad_sch

backup

1S LiPo backup: charger, boost, ideal-diode ORing into both 5 V rails

File: backup.kicad_sch

test

Test points (bench only): incoming, outgoing, measured

File: test.kicad_sch

Generated by hardware/gen/make_sch.py from gen/design.py — do not hand-edit
Dual-channel steering torque-overlay unit, Honda Insight EPS

Sheet: /
File: icarus.kicad_sch

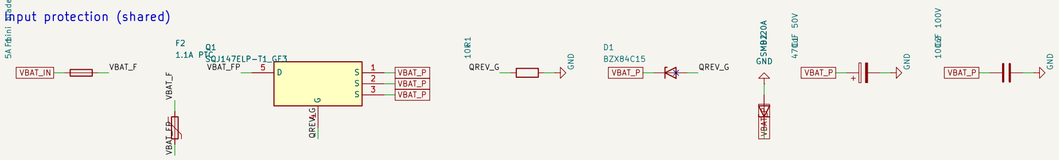
Title: Icarus — top level

Size: A3	Date: 2026-11-01	**Rev: 0.5**
KiCad E.D.A. 10.0.0		Id: 1/10

Power: input protection and two independent supply chains

Everything after VBAT_P is duplicated per channel (SM-16): buck -> LDO -> supervisors.

Input protection (shared)



Supply chain A: LMR38020-Q1 5.0 V buck, TPS7B81-Q1 3.3 V LDO, TPS3702-Q1 5 V window



Supply chain B: LMR38020-Q1 5.0 V buck, TPS7B81-Q1 3.3 V LDO, TPS3702-Q1 5 V window

Indicators: rail present (green, ~1.4 mA each)

Power flags (rails driven by regulators / the protection FET)

Generated by hardware/gen/make_sch.py from gen/design.py - do not hand-edit
Dual-channel steering torque-overlay unit, Honda Insight EPS

Sheet: /power/
File: power.kicad_sch

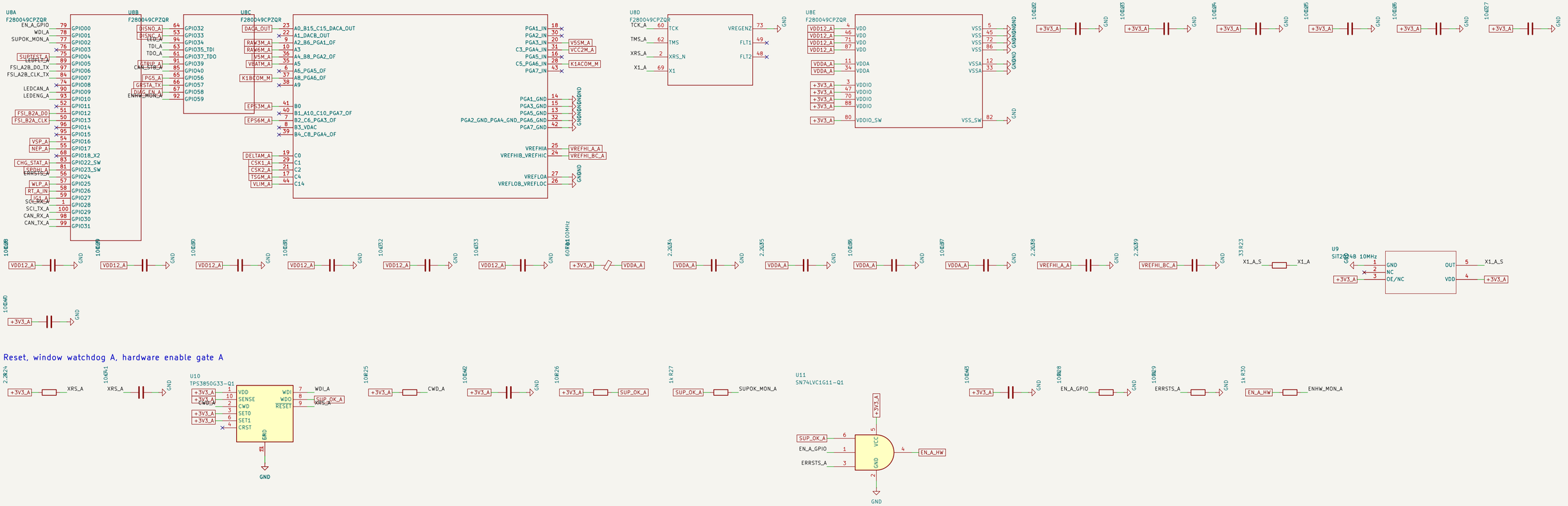
Title: Icarus — Power: input protection and two independent supply cha

Size: A2	Date: 2026-11-01	**Rev: 0.5**
KiCad E.D.A. 10.0.0		Id: 2/10

CH-A: F280049C-Q1 control (computes and drives the overlay)

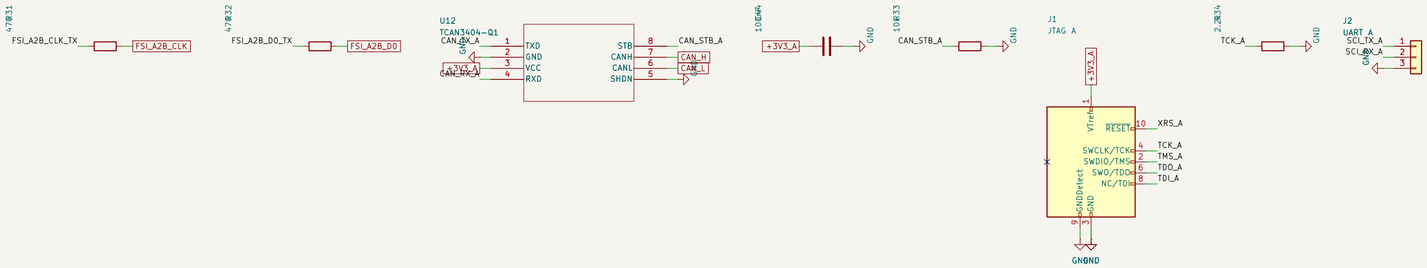
Pin mux from SPRS945 Table 5-6. Window watchdog TPS3850-Q1: CWD 10k to VDD, SET0 = SET1 = 1 -> 1.85 ms open / 11 ms timeout, service every ~5 ms.

MCU A

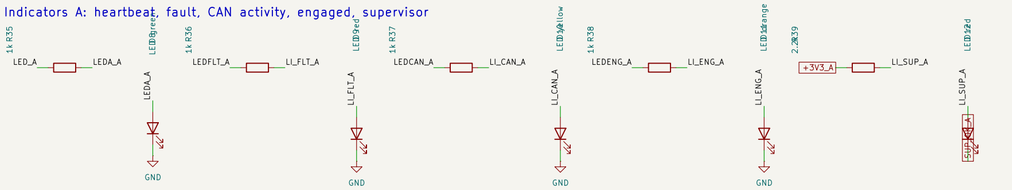


Reset, window watchdog A, hardware enable gate A

FSI to CH-B, CAN, debug



Indicators A: heartbeat, fault, CAN activity, engaged, supervisor



Generated by hardware/gen/make_sch.py from gen/design.py — do not hand-edit
Dual-channel steering torque-overlay unit, Honda Insight EPS

Sheet: /ch_a/
File: ch_a.kicad_sch

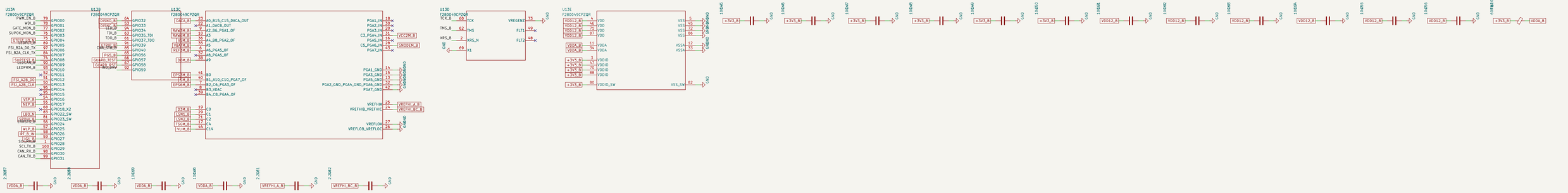
Title: Icarus — CH-A: F280049C-Q1 control (computes and drives the o

Size: A2	Date: 2026-11-01	Rev: 0.5
KiCad E.D.A. 10.0.0		Id: 3/10

CH-B: F280049C-Q1 monitor (checks; owns the low-side coil enable)

Pin mux from SPRS945 Table 5-6. Window watchdog TPS3850-Q1: CWD 10k to VDD, SET0 = SET1 = 1 -> 1.85 ms open / 11 ms timeout, service every ~5 ms.

MCU B

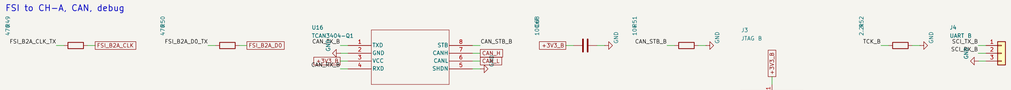


Reset, window watchdog B, hardware enable gate B

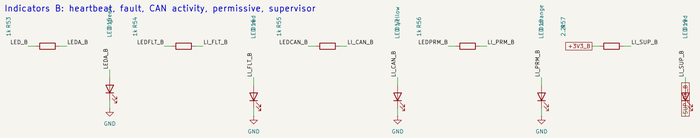
Dynamic enable B: charge pump, PWM only (SM-04)

Driver indicator output (low side)

FSI to CH-A, CAN, debug



Indicators B: heartbeat, fault, CAN activity, permissive, supervisor



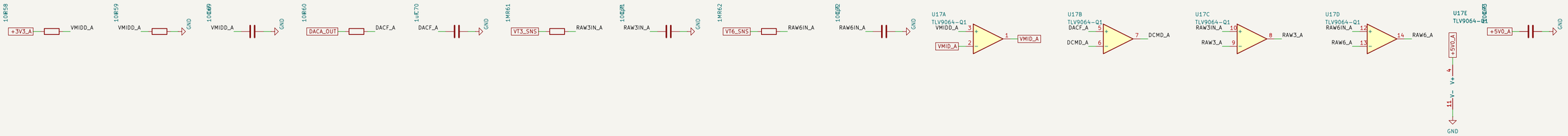
Generated by hardware/gen/make_sch.py from gen/design.py — do not hand-edit
Dual-channel steering torque-overlay unit, Honda Insight EPS

Sheet: /ch_b/	
File: ch_b.kicad_sch	
Title: Icarus — CH-B: F280049C-Q1 monitor (checks; owns the low-side	
Size: A1	Date: 2026-11-01
Rev: 0.5	
KiCad E.D.A. 10.0.0	Id: 4/10

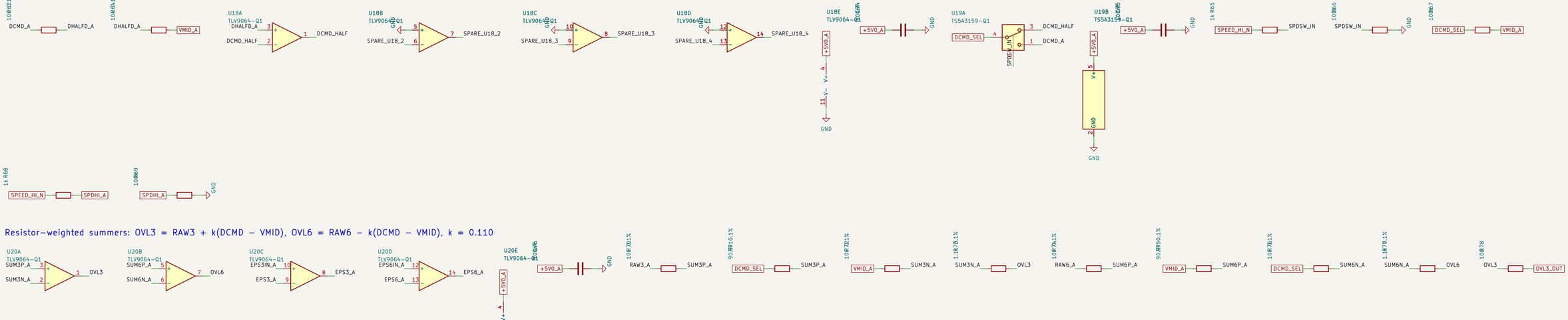
Overlay path (CH-A analog, +5V0_A)

EPS gets OVL = RAW + (DELTA − VMID). DELTA = VMID + k(DAC − VMID), k = 10k/92k = 0.109 -> Delta_max = +/−0.18 V (TBD-DATA, SM-05). One DAC + inverter makes VT6 = −Delta (SM-06). Summing resistors 0.1 %.

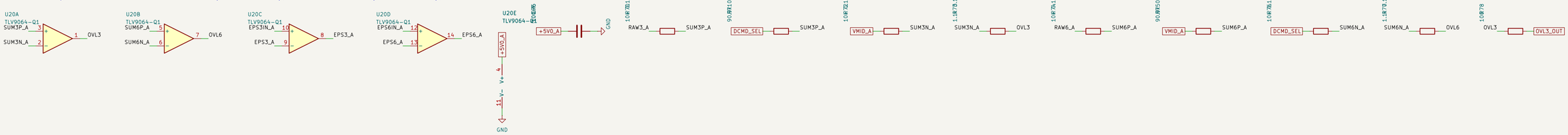
Reference, Delta command and 1 MOhm sensor-side taps



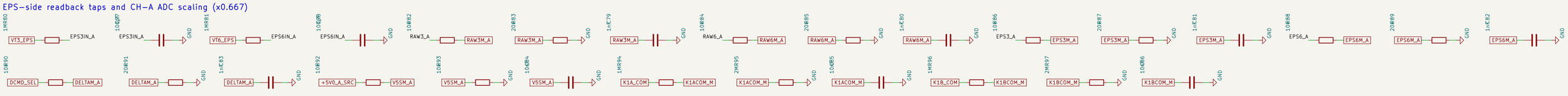
Speed-dependent authority (SM-23): half-amplitude command above v_switch



Resistor-weighted summers: OVL3 = RAW3 + k(DCMD − VMID), OVL6 = RAW6 − k(DCMD − VMID), k = 0.110



EPS-side readback taps and CH-A ADC scaling (x0.667)



Generated by hardware/gen/make_sch.py from gen/design.py — do not hand-edit
Dual-channel steering torque-overlay unit, Honda Insight EPS

Sheet: /overlay/	
File: overlay.kicad_sch	
Title: Icarus — Overlay path (CH-A analog, +5V0_A)	
Size: A2	Date: 2026-11-01
KiCad E.D.A. 10.0.0	**Rev: 0.5**
	Id: 5/10

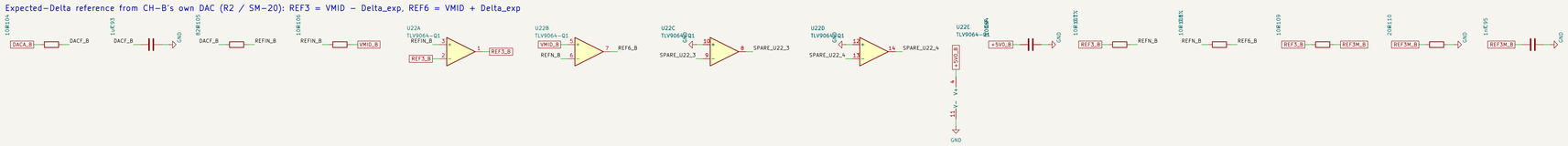
CH-C analog guard (no software, +5V0_B)

Trips when the overlay actually applied (VT_EPS − VT_SNS) differs from CH-B's own expected Delta by > 0.25 V on either line (R2 / SM-20), or the VT3+VT6 sum moves > 0.10 V (TBD-DATA). A trip latches until key-off and cuts every coil (SM-15). GUARD_TEST forces a trip for the key-on self-test (SM-17). VMID_B and the ladder are ratiometric to +3V3_B, the CH-B DAC reference.

Buffers (1 MOhm from the car side) and VMID_B



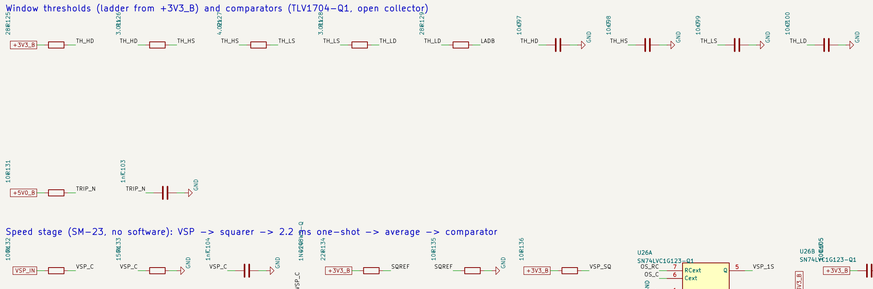
Expected-Delta reference from CH-B's own DAC (R2 / SM-20): REF3 = VMID − Delta_exp, REF6 = VMID + Delta_exp



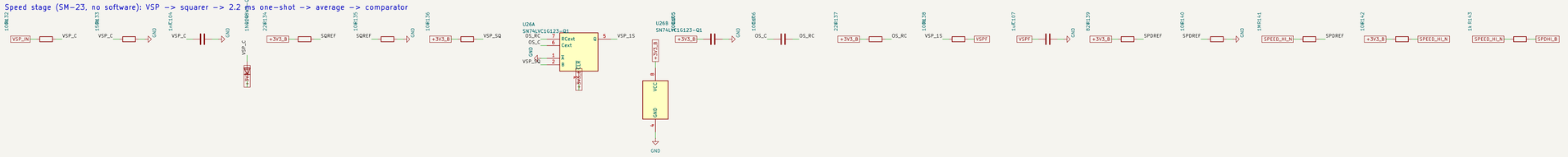
Difference amps: D3 = EPS3 − RAW3 + REF3 (= VMID when the applied Delta matches the expected), D6 likewise, S = (EPS3+EPS6) − (RAW3+RAW6) + VMID



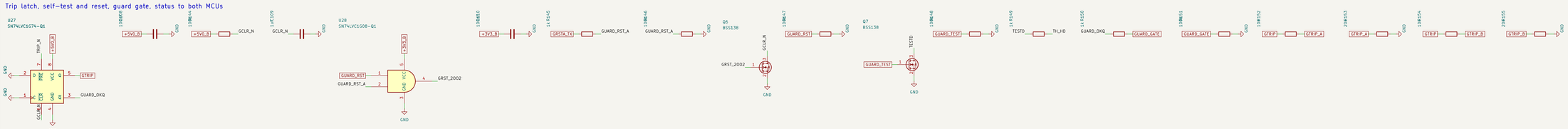
Window thresholds (ladder from +3V3_B) and comparators (TLV1704-Q1, open collector)



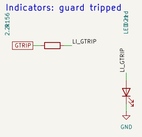
Speed stage (SM-23, no software): VSP -> squarer -> 2.2 ms one-shot -> average -> comparator



Trip latch, self-test and reset, guard gate, status to both MCUs



Indicators: guard tripped



CH-B ADC scaling (x0.667)



Generated by hardware/gen/make_sch.py from gen/design.py — do not hand-edit
Dual-channel steering torque-overlay unit, Honda Insight EPS

Sheet: /guard/
File: guard.kicad_sch

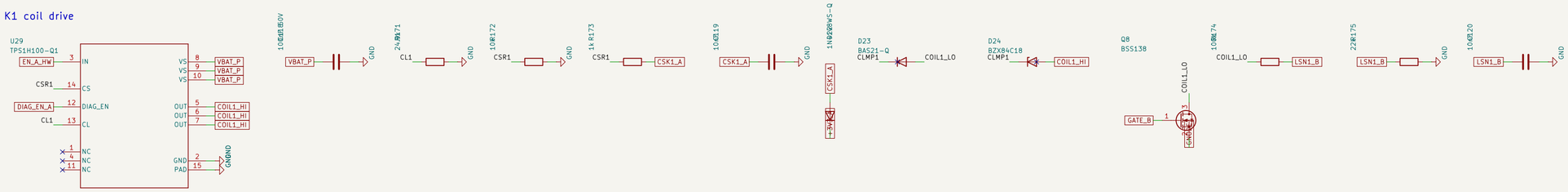
Title: Icarus — CH-C analog guard (no software, +5V0_B)

Size: A1 | Date: 2026-11-01 | **Rev: 0.5**
KiCad E.D.A. 10.0.0 | Id: 6/10

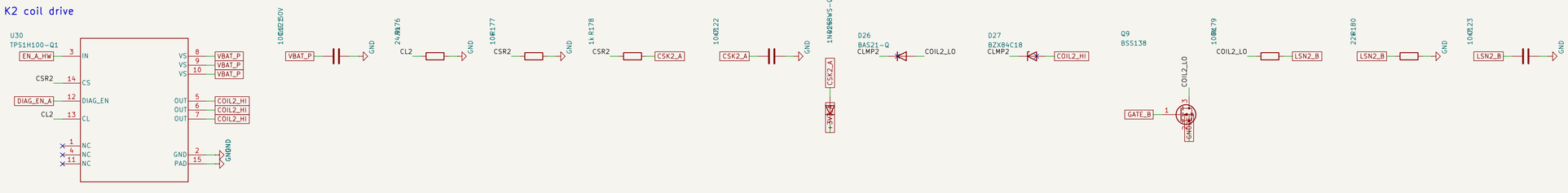
Signal relays K1/K2 (cascaded) and cross-driven coils

EPS sees the overlay only if K1 AND K2 are energized (SM-14). Each coil: high side from CH-A (TPS1H100-Q1), low side from CH-B (charge-pumped gate) and the CH-C guard FET in series.

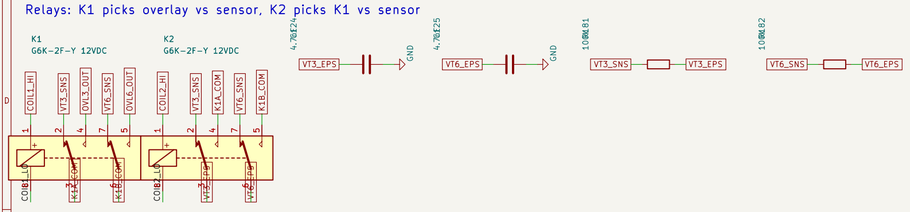
K1 coil drive



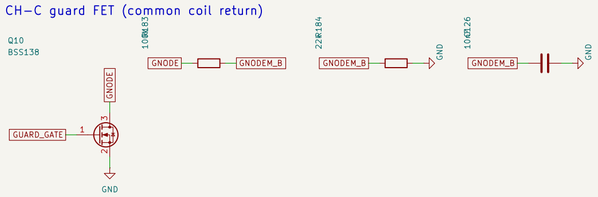
K2 coil drive



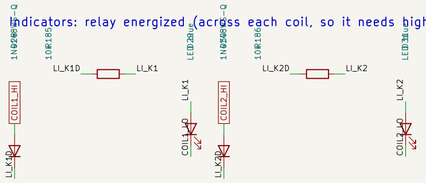
Relays: K1 picks overlay vs sensor, K2 picks K1 vs sensor



CH-C guard FET (common coil return)



Indicators: relay energized (across each coil, so it needs high side AND both low sides)



Generated by hardware/gen/make_sch.py from gen/design.py — do not hand-edit
Dual-channel steering torque-overlay unit, Honda Insight EPS

Sheet: /relays/
File: relays.kicad_sch

Title: Icarus — Signal relays K1/K2 (cascaded) and cross-driven coils

Size: A2 | Date: 2026-11-01 | **Rev: 0.5**
KiCad E.D.A. 10.0.0 | Id: 7/10

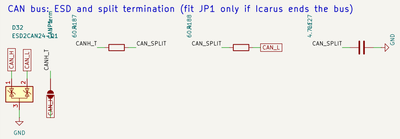
Vehicle connector, CAN bus, inputs and monitors

Every car-facing input has its own divider per channel (no shared resistor between CH-A and CH-B). Torque-sensor and Vcc2 taps are 1 MOhm (FSR-14).

Vehicle connector (Y-harness at EPS connector B; part TBD, TE MCP family as throttle J2)



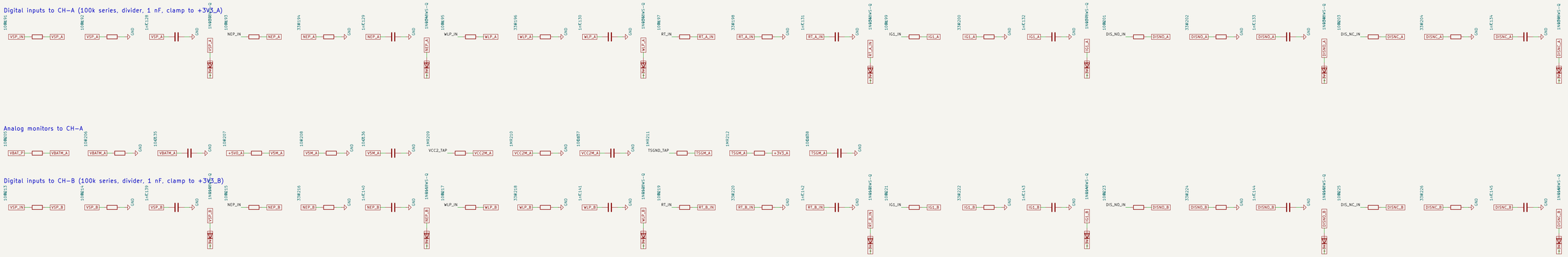
CAN bus: ESD and split termination (fit JP1 only if Icarus ends the bus)



Disengage switch wetting (NO + NC contacts to GND)



Digital inputs to CH-A (100k series, divider, 1 nF, clamp to +3V3_A)



Analog monitors to CH-A

Digital inputs to CH-B (100k series, divider, 1 nF, clamp to +3V3_B)

Analog monitors to CH-B



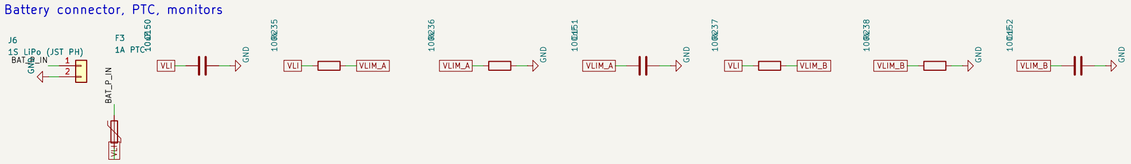
Generated by hardware/gen/make_sch.py from gen/design.py — do not hand-edit
Dual-channel steering torque-overlay unit, Honda Insight EPS

Sheet: /io/	
File: io.kicad_sch	
Title: Icarus — Vehicle connector, CAN bus, inputs and monitors	
Size: A1	Date: 2026-11-01
KiCad E.D.A. 10.0.0	Rev: 0.5
	Id: 8/10

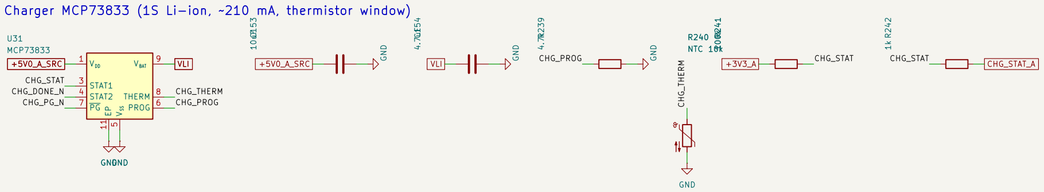
1S LiPo backup: charger, boost, ideal-diode ORing into both 5 V rails

Keeps both MCUs, CAN and the analog guard alive through cranking / IMA-restart dips so a dropout is reported (SG4) and recovery is clean. The relay coils are NOT backed up on purpose: 12 V loss = pass-through. Charger input is +5V0_A_SRC (ahead of the ideal diode), so backup power cannot loop back into the charger. Pack must have its own PCM; charge window 0-45 C via the on-board NTC.

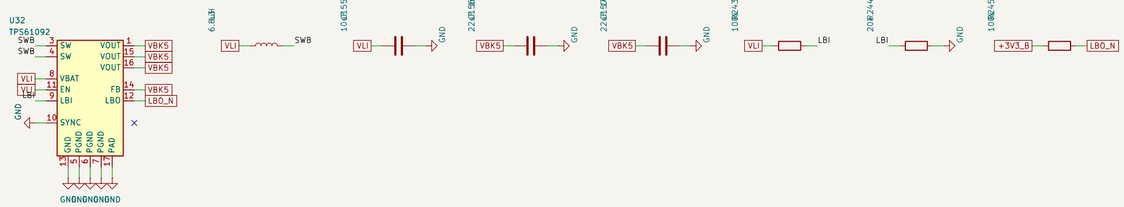
Battery connector, PTC, monitors



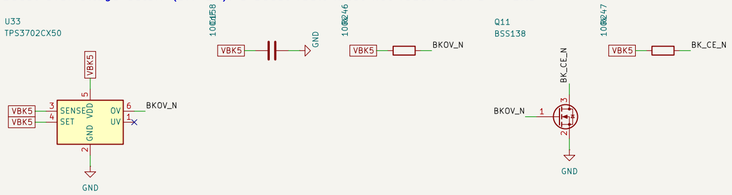
Charger MCP73833 (1S Li-ion, ~210 mA, thermistor window)



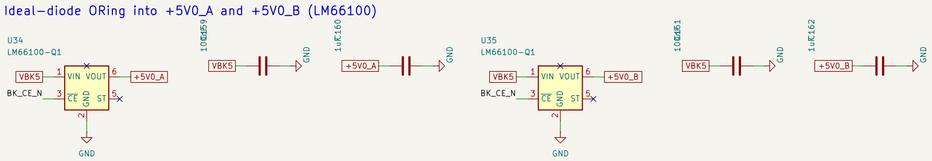
Boost TPS61092 (fixed 5.0 V) and low-battery flag



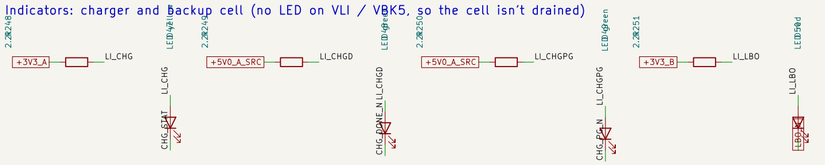
Boost overvoltage cutoff (SM-24): a boost fault must not reach both 5 V rails



Ideal-diode ORing into +5V0_A and +5V0_B (LM66100)



Indicators: charger and backup cell (no LED on VLI / VBK5, so the cell isn't drained)



Generated by hardware/gen/make_sch.py from gen/design.py - do not hand-edit
Dual-channel steering torque-overlay unit, Honda Insight EPS

Sheet: /backup/	
File: backup.kicad_sch	
Title: Icarus - 1S LiPo backup: charger, boost, ideal-diode ORing into	
Size: A2	Date: 2026-11-01
KiCad E.D.A. 10.0.0	**Rev: 0.5**
	Id: 9/10

Test points (bench only): incoming, outgoing, measured

1.5 mm SMD pads, no parts fitted. Each pad sits on its net (a probe point, never in series), in three labelled clusters on the PCB: signals coming in from the car, signals Icarus drives out, and internal measurement nodes. Each cluster has its own GND pad for the scope ground. Torque-line pads are for the bench; don't probe VT3/VT6 while driving.

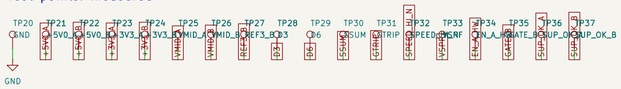
Test points: incoming



Test points: outgoing



Test points: measured



Generated by hardware/gen/make_sch.py from gen/design.py — do not hand-edit
Dual-channel steering torque-overlay unit, Honda Insight EPS

Sheet: /test/
File: test.kicad_sch

Title: Icarus — Test points (bench only): incoming, outgoing, measured

Size: A2	Date: 2026-11-01	**Rev: 0.5**
KiCad E.D.A. 10.0.0		Id: 10/10